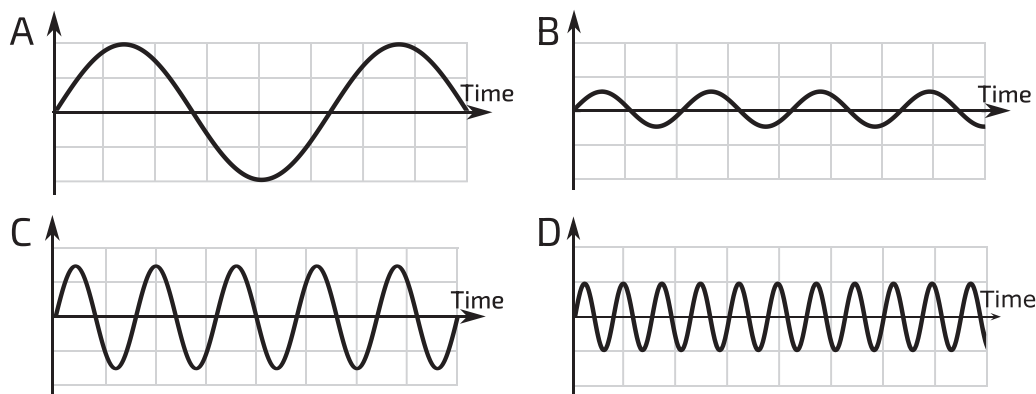


Sound and Waveforms Practice

- 1 Look at the waves shown below and answer the questions.



- (a) Rank the waves from loudest to quietest: _____
 (b) Rank the waves from highest to lowest pitch: _____
 (c) Which wave in Figure 1 has the shortest time period? _____

- 2 Which of the waves from question 1 has the greatest frequency? _____

- 3 Complete the sentences using **Hz, kHz, 3, 40 and 1000**. You may use each more than once.

If a frequency is very high, we might measure it in _____, which are each equal to _____.

For example, a beeping smoke alarm has a frequency of roughly _____, while an open E note on a bass guitar is roughly _____.

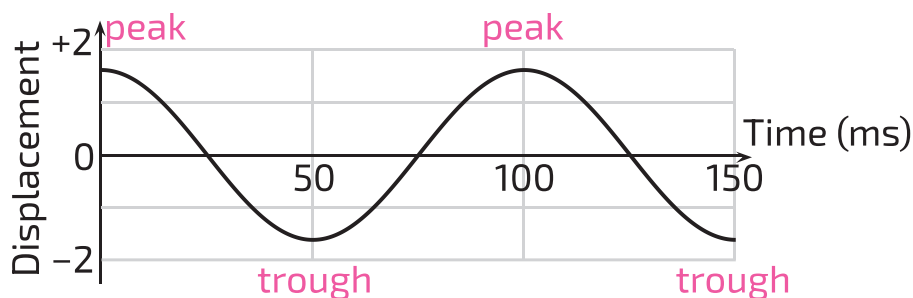
- 4 Write the following times in milliseconds.

(a) 0.007 s (b) 0.301 s (c) 0.05 s

- 5 Write the following times in seconds.

(a) 5000 ms (b) 9 ms (c) 40.1 ms

- 6 What is the time period, in seconds, of this wave?



- 7 (a) A wave has a time period of 0.2 s. Work out how many times it oscillates each second using an equation.

$$\boxed{1} \div \boxed{} = \boxed{}$$

- (b) What is the frequency of a wave if its time period is 0.2 s?

- (c) A wave has a time period of 0.008 s. Work out its frequency using an equation.

$$\frac{1}{\boxed{1}} \div \text{time period (s)} = \text{frequency (Hz)}$$

$$\boxed{1} \div \boxed{} = \boxed{}$$

- (d) Work out the frequency of a wave with a time period of 0.08 s using an equation.

$$\frac{1}{\boxed{1}} \div \text{time period (s)} = \text{frequency (Hz)}$$

$$\boxed{1} \div \boxed{} = \boxed{}$$

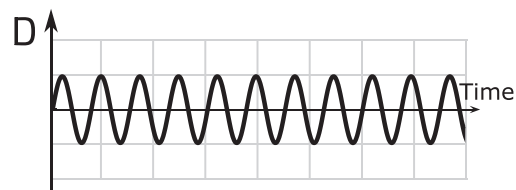
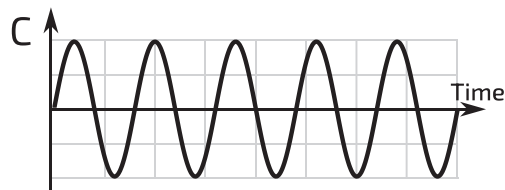
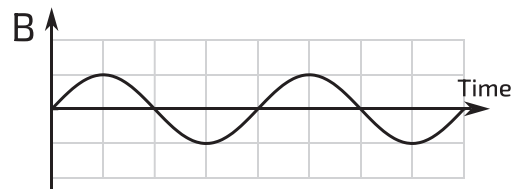
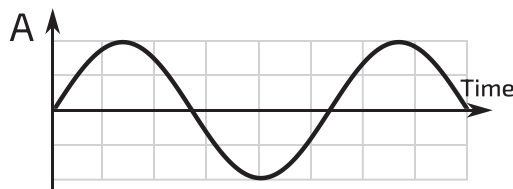
- 8 A wave has a time period of 800 ms. Calculate its frequency.

- 9 What is the frequency of the wave in question 6?

- 10 Calculate the frequencies in kHz of waves with the following time periods.

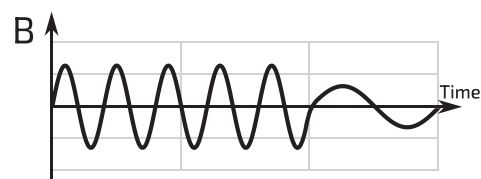
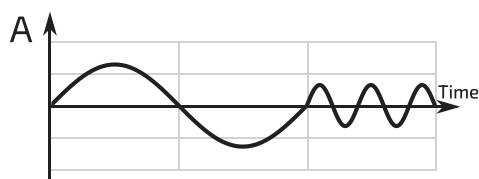
- (a) 25 ms (b) 200 ms (c) 0.4 ms (d) 0.001 ms

- 11 Each graph represents the sound wave for one of the situations in the table below. Match the situation to the graph.



- (a) The buzzing from a mosquito's wings (c) The booming of thunder after lightning
(b) A singer hitting a high note at a festival (d) A quiet bass guitar backing track

- 12 A guitarist is tuning her guitar. She plays a note and then quickly tightens the string, increasing its pitch.



Which of the graphs in Figure 12 most accurately represents the sound that the guitar makes?